

ANISH AMBRETH

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LinkedIn ↗ ◊ Website ↗

SUMMARY

Researcher with strong foundations in probability, statistics and machine learning. Experienced in Federated Learning, Agentic AI and predictive modeling.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE

Aug 2024 – May 2026

M.Sc. in Machine Learning

GPA: 3.95/4.00

- Fully funded scholarship from the UAE government
- Thesis: Adversarial Robustness in Federated Learning

Indian Institute of Technology (IIT) Hyderabad, India

Aug 2020 – May 2024

B.Tech in Mathematics & Computing and Engineering Sciences

GPA: 3.42/4.00

PUBLICATIONS

1) **Anish, A.**, Naveen, k., Mohsen, G. (2026). Client-Level Defense Placement for Adversarially Robust Federated Reinforcement Learning. *Under review at TMLR'26*

2) Tameem, B.* , **Anish, A.***, Nils, L. (2026). Collaborative Threshold Watermarking. *Under review at ICML'26. (* equal contribution).*

3) **Anish, A.**, Naveen, k., Mohsen, G. (2026). Strategic Client-Level Defense Placement in Mobile Federated Learning. *Under review at Mobihoc'26*

4) **Anish, A.**, Naveen, k., Mohsen, G. (2026). OTPA: Optimal Transport-Based Model Alignment Poisoning in Mobile and Edge Federated Learning. *Under review at Mobihoc'26*

TECHNICAL SKILLS AND COURSEWORK

- **Core:** Python, NumPy, pandas, scikit-learn, PyTorch
- **Statistics & Math:** Probability theory, statistical inference, regression, hypothesis testing, optimization, numerical methods
- **Tools:** Jupyter, Git, Docker, SQL

PROFESSIONAL EXPERIENCE

ML Quantitative Research Intern – Vatic Labs (UAE)

Jan 2026 – Apr 2026

- Developed a research pipeline for alpha factor mining from raw order book data using a multi-agent system to generate candidate predictive signals.
- Incorporated generated factors into predictive time-series models, including TCNs and LSTMs, and tested their impact on historical forecasting performance.
- Supported the buildout of an end-to-end systematic trading research workflow, from signal generation to model evaluation, with future extension toward execution.

ML Quantitative Research Intern – Standard Chartered (UAE)

Jul 2025 – Aug 2025

- Predictive models for FX rate dynamics (PKR, INR, NPR), focusing on robustness.
- Engineered and evaluated feature sets derived from market and competitor pricing data to support return-sensitive pricing decisions.

Research Intern – Dubai Health (UAE)

Apr 2025 – Jul 2025

- Developed and deployed a **lightweight snoring detection** algorithm on ESP Sense for sleep monitoring, using signal processing and ML to infer events directly on-device.
- Achieved **99% reduction** in storage/processing and **50% lower** power consumption compared to phone-based app approaches, enabling scalable, cost-effective deployment.
- Worked with clinicians and engineers to **align model behavior with medical requirements**, presenting findings in an accessible, non-technical way.

Data Science Intern – ADNOC (UAE)

Jan 2025 – Mar 2025

- Built forecasting and scoring models on heterogeneous time-series and textual data to rank emerging technologies.
- Designed evaluation metrics and validation schemes to reduce noise and bias in model-driven rankings.
- Automated feature extraction and scoring pipelines, reducing manual analysis time by 90%.

Machine Learning Intern – Proshort (India)

Jun 2024 – Aug 2024

- Fine-tuned and quantized LLMs (LLaMA-3-8B, Phi3-mini-128k-Instruct) in PyTorch, tracking experiments with Weights & Biases to systematically optimize quality and latency.
- Deployed models using **Docker** and **FastAPI**, integrating RAG pipelines to improve response quality and accuracy by **30%**, evaluated with ROUGE-2 and human feedback.
- Designed and optimized **LangGraph-based workflows** and user session management, improving conversation continuity and robustness in production.

Software Developer Intern – Denso International (India)

May 2023 – Jul 2023

- Built ECU data analysis software for Suzuki Motors to process and visualize error codes, **reducing manual analysis effort by 70%**.
- Developed an application for **ADAS data collection**, integrating camera and GPS streams for synchronized viewing and diagnostics, improving debugging efficiency for engineering teams.

Research Intern – Santa Fe Research (India)

May 2022 – Sep 2022

- Optimized vehicle routing using **OR-Tools** with time, capacity, and dimensional constraints, integrating 3D bin packing to improve logistical efficiency.
- Formulated and solved operations research problems with a focus on **scalable, algorithmic decision-making** in real-world routing scenarios.

PROJECTS

Quantitative Reasoning for Text-to-Image Generation

- Curated a benchmark dataset to evaluate **quantitative reasoning** (counting, proportionality, fractions) in text-to-image diffusion models.
- Fine-tuned Stable Diffusion XL with **LoRA** and **DPO**, improving quantitative faithfulness in generated images; analyzed performance across multiple prompt categories.

Quantitative Analysis of Time Series Methods Performance

- Modeled and forecasted volatility and price dynamics of Gold, S&P 500, and Bitcoin using ARIMA, LSTM, XGBoost, and Prophet.
- Engineered lagged, rolling, and volatility-based features; evaluated models using MAE, RMSE, and stability across regimes.
- Analyzed bias-variance tradeoffs and robustness of models under non-stationary conditions.

Study of Differential Equations in Fluid Dynamics and Solute Dispersion

- Developed and analyzed partial differential equation models for fluid flow and solute dispersion, strengthening **mathematical modeling and numerical analysis** foundations.

POSITIONS OF RESPONSIBILITY

Graduate Assistant – MBZUAI (UAE)

Aug 2025 – Oct 2025

- Assisted faculty with course and lab delivery, including content preparation, assignment support, and grading logistics.
- Helped organize and evaluate examinations, ensuring consistency and fairness across students.

Basketball Team Lead

Aug 2020 – Present

- Team Lead** | Led IIT Hyderabad team to nationals, securing 4th place twice out of 23 teams.
- Team Lead** | Leading MBZUAI team in competitions, coordinating training, strategy, and team dynamics.

VOLUNTEER EXPERIENCE

Well-being Coordinator

Aug 2024 – Present

Cultural Events Coordinator

Oct 2024 – Nov 2024

- Well-being Coordinator** – Hosted events with 300+ attendees, fostering student well-being and engagement.
- Cultural Events Coordinator** – Organized Diwali celebration for 200+ international students, showcasing culture and building community.